

CS & AI • EXPLAINED

How does AI like ChatGPT actually work?

No hype, no jargon — what's really happening inside the box.

THE HOOK

You type "I am going ho..." and your phone guesses "home".

ChatGPT does the same thing — just a billion times bigger.

In 5 minutes, you'll actually understand the "magic". 🧠



TODAY'S QUESTION

How does it **decide** what to say
—
and does it actually **know**
things?

IDEA 1 / 4 — IT'S A PREDICTION MACHINE

It predicts the **next word**.

"The cat sat on the ___"



- It doesn't **look up** an answer.
- It picks the **most likely** next word...
- ...then repeats, again and again.

It read a **huge** amount of text.

- Trained on **billions** of sentences — books, websites, code.
- It learned **patterns** of language...
- ...**not** a saved list of facts it copies.



books



websites



conversations



code

Like a student who read millions of pages and absorbed **how language flows**.

One piece at a time.

• • •

```
# prompt
```

```
Why is the sky blue?
```

```
# it generates token by token, feeding each back in →
```

```
The          → The sky          → The sky looks
```

```
→ The sky looks blue    → The sky looks blue because ...
```

Each word is a **token**. Pick one, add it, predict the next. Repeat.

It weighs **every word** you wrote.

“I left my phone in the car, can you help me find it?”

- To choose the next word, it asks: **which earlier words matter most?**
- Here, **phone** + **car** shape the answer — not “the” or “my”.
- This “pay attention to context” trick is what made modern AI work.

WHY IT MATTERS

Same trick → essays, code, emails, answers.

- One simple idea — predict the next token — scales to almost any text task.
- That's why a chatbot can debug code *and* write a poem.

BUT REMEMBER

It's predicting, not understanding.

COMMON MYTH — "YEH GALAT SAMAJHTE HAIN"

X MYTH

"ChatGPT **knows** the truth and looks it up."

✓ REALITY

It predicts the most **likely** words. So it can sound confident and still be **wrong** — that's a hallucination. Always verify.

RECAP — 2 THINGS TO REMEMBER

- 1 AI like ChatGPT = **next-word prediction**, at massive scale.
- 2 It doesn't think or know — it **guesses patterns**. So check the important facts.

Now you understand the "magic". 

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Next question: [What is an algorithm?](#) — the samosa-queue secret 🥟